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Introducing a Customized Workflow to Analyze DFIT-FBA Post-Flowback Shut-In (Rebound) Data

S. Haqparast, Department of Chemical & Petroleum Engineering, University of Calgary, Calgary, AB; C. R. Clarkson, Department of Earth, Energy, and Environment, University of Calgary, Calgary, AB

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Abstract

DFIT-FBA (FBA = flowback analysis) offers a faster alternative to traditional diagnostic fracture injection tests (DFITs) for determining key parameters used in hydraulic fracturing design. While a method exists for estimating reservoir pressure from the flowback period of DFIT-FBA, this method could be inaccurate for certain cases. In this study, shut-in data (also called rebound) recorded after the flowback period of the DFIT-FBA test were analyzed to provide an independent estimate of reservoir pressure.

An extensive DFIT-FBA post-flowback shut-in test was simulated and analyzed to interpret all pressure signatures. This detailed investigation provided valuable insights into physical mechanisms occurring during the post-flowback shut-in period of a DFIT-FBA test. Based on these findings, a customized analysis workflow was developed and then utilized to estimate reservoir pressure from post-flowback shut-in data for nine field cases. These results were then compared with the results obtained from the analysis of the flowback period of DFIT-FBA.

Key observations from the simulation study include: 1) linear flow is the only reservoir flow regime observed during the post-DFIT-FBA shut-in data of the simulated case; 2) reservoir pressures estimated from the shut-in test are always higher than the actual reservoir pressure even after an extended shut-in period. The major takeaways from analyzing field examples include: 1) shut-in pressures recorded after a DFIT-FBA test reach a maximum, and will decline if pressure is monitored long enough; 2) more accurate reservoir pressure estimates will be obtained if the test is extended far beyond the point where maximum pressure is observed; 3) if the minimum horizontal stress and reservoir pressure are similar in magnitude, and/or occur close in time, the reservoir pressure signature during the flowback period of a DFIT-FBA could be masked by the fracture closure event. In such cases, the reservoir pressure estimate obtained from the analysis of DFIT-FBA flowback data should be used with caution, unless confirmed through independent estimates, such as DFIT-FBA post-flowback shut-in data.

Introduction

Estimating reservoir pressure in low-permeability ('unconventional') reservoirs is challenging but critical for field development planning and reserves estimation. One commonly used approach is to perform a diagnostic fracture injection test (DFIT) through a vertical well or at the toe of a horizontal well. The conventional DFIT involves injecting fluid at high pressure to initiate and propagate a small hydraulic fracture, followed by an immediate shut-in. The subsequent pressure decline is then analyzed to derive key reservoir properties including reservoir pressure. Accurate estimation of reservoir pressure using this method requires extended monitoring to capture data beyond the point when the fracture is dynamically closed (Nolte, 1997; Craig and Blasingame, 2006). However, the time needed to reach this phase can range from several hours to even months.

An alternative DFIT design is the flowback DFIT (Nolte 1979; Nolte and Smith 1981; Smith 1985). This approach involves flowing back the injected fluid through a surface choke after injection stops, with the flowing pressure recording during flowback used to infer the minimum in-situ stress. The flowback DFIT was introduced to accelerate fracture closure and obtain a more rapid estimate of minimum in-situ stress compared to the conventional DFIT. Nolte recommended a flowback rate of approximately one-sixth to one-quarter of the injection rate. Although this method effectively accelerates fracture closure, a key drawback is the potential loss of wellbore-reservoir connectivity at high flowback rates, which can lead to pressure data being primarily influenced by wellbore unloading rather than reservoir responses. To address this limitation, Zanganeh et al. (2020a) introduced an alternative approach known as DFIT-FBA, where FBA refers to flowback analysis. This method builds on the principles of flowback analysis performed on early-time production data gathered immediately after the hydraulic fracture stimulation of a well (Clarkson and Williams-Kovacs, 2019). For the DFIT-FBA procedure, Zanganeh et al. (2020a) proposed flowing back at a lower rate (generally 2-5% of injection rate) to maintain connectivity, enabling the acquisition of after-closure reservoir information while also accelerating closure. They also suggested that flowback rate and pressures could be analyzed using rate-transient analysis (RTA). A properly designed, implemented, and analyzed DFIT-FBA test can deliver the same critical parameters as a traditional DFIT but within a much shorter timeframe, typically a few hours in low-permeability systems, compared to days or weeks with the conventional DFIT method (Clarkson et al. 2025). The increased speed of DFIT-FBA opens up new applications for DFITs, including rapid target zone selection for horizontal wells during time-sensitive exploration programs (Zanganeh et al. 2020b), pressure and permeability evaluation along horizontal laterals (Zeinabady et al. 2022b), assessment of induced seismicity risks (Zeinabady et al. 2023a), and optimization of fracturing fluid selection (Haqparast et al. 2024).

For reservoir pressure estimation using DFIT-FBA, Zanganeh et al. (2020a) suggested selecting the pressure corresponding to the start of the second unit slope line occurring on a log-log plot of the rate-normalized pressure derivative (RNP') with respect to material balance time (MBT) (Figure 1). RNP, RNP' and MBT are defined as follows:

$$RNP(\Delta t) = \frac{p_{ref} - p_f(\Delta t)}{q_{fb}(\Delta t)} \quad (1)$$

$$RNP'(\Delta t) = \frac{dRNP(\Delta t)}{d \ln(t_{MB}(\Delta t))} \quad (2)$$

$$t_{MB}(\Delta t) = \frac{V_{fb}(\Delta t)}{q_{fb}(\Delta t)} \quad (3)$$

where p_{ref} is the pressure at the start of flowback, q_{fb} is the flowback rate, and $V_{fb}(\Delta t)$ is the cumulative flowback volume. They suggested that the second unit slope line (Figure 1) indicates boundary-dominated

flow in the fluid bank region (FBR) adjacent to the fracture in which the pressure has been elevated during the injection, shut-in, and early flowback period due to fluid leakoff from the fracture. Zeinabady (2023) demonstrated through a set of simulation cases that this method overestimates reservoir pressure. He also showed that the accuracy of the estimated pressure decreased in higher permeability cases, where more fluid leaked into the reservoir and elevated the FBR pressure to a greater extent. It should be noted that reservoir pressure is used to estimate reservoir permeability from the method presented by Zeinabady et al. (2022a). Therefore, in addition to being an important parameter to obtain from DFIT-FBA for other purposes, reservoir pressure affects the reservoir permeability estimate.

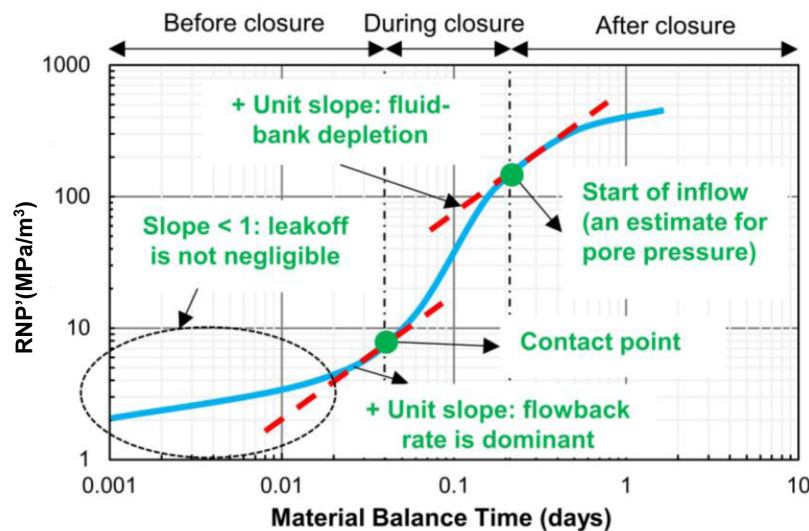


Figure 1—Typical log-log plot of rate-normalized pressure derivative (RNP') with respect to material balance time (MBT) for a DFIT-FBA test. Fracture closure begins with the contact point where RNP' starts deviating from the first unit slope line (After Zeinabady 2023e).

A challenge in estimating reservoir pressure with the current DFIT-FBA analysis workflow is that the second unit slope line becomes visible only after the fracture closure signature has been completed. The start of fracture closure is marked by a deviation from the first unit slope line (indicating boundary-dominated flow in the fracture), where the total system stiffness increases due to the contact between fracture walls. This point is known as the contact point (Figure 1) from which the contact pressure is estimated. During fracture closure, the pressure data is affected by changing system stiffness, and as such, cannot and should not be used to interpret reservoir signatures. Consequently, the reservoir pressure signature, represented by the start of the second unit slope line, may be masked if the minimum horizontal stress and reservoir pressure are close in magnitude and/or occur in close proximity in time.

Another issue with estimating reservoir pressure using the DFIT-FBA method can occur when the test is run using a constant rate setting. At the end of the flowback period, it may not be possible to hold the rate constant. As a result, the well will start to flow under constant pressure conditions. This can create a unit-slope line in the data, which might be mistakenly interpreted as the beginning of boundary-dominated flow in the fluid bank region (FBR). Similarly, if DFIT-FBA is conducted using a constant choke setting (which keeps surface pressure constant), the use of material balance time may create an artificial unit-slope line due to a rapid decline in rates at the end of the flowback period (Clarkson et al. 2025). In short, it may be difficult to identify a true unit slope at the end of the flowback period.

Finally, as noted above, Zanganeh et al. (2020a) suggested a range of flowback rates (rule-of-thumb: 2-5% of the injection rate) to ensure that connectivity between the fracture and the wellbore is maintained while making it possible to analyze the data using rate-transient analysis (RTA) methods. Later, Zeinabady et al. (2023b) introduced an analytical method and procedure to obtain an optimized injection volume and

flowback rate to increase the chance of a successful DFIT-FBA test. They noted that the test can fail if the flowback rate is too high or too low, resulting in an unidentifiable closure signature. In certain instances, such as in naturally-fractured reservoirs, unexpectedly high fluid leakoff can occur into the reservoir. In such cases, the fracture may close off (pinch), and the connection between the fracture and the well may be lost. When this happens, wellbore unloading may dominate the flowback signature, making it impossible to estimate reservoir pressure. A similar issue can occur in cases with high near-wellbore frictions, where a large pressure drop at the beginning of the flowback period can mask the fracture closure signature.

The challenges mentioned above have inspired an investigation of alternative methods to estimate reservoir pressure from the DFIT-FBA test. One such source is shut-in (buildup or rebound) data recorded after the flowback period of a flowback-DFIT. However, limited comprehensive research has been conducted on rebound data. [Plahn et al. \(1997\)](#) conducted a series of simulations to explore the relationship between closure pressure and the maximum pressure recorded during the shut-in. They found that maximum pressure tends to approach closure pressure values obtained from the flowback DFIT test. However, many DFIT-FBA tests analyzed by the authors have demonstrated that the last recorded pressure obtained during the post-flowback shut-in period is significantly lower than the closure pressure. Other studies have also attempted to use rebound data, but their objective was primarily to constrain or validate the closure pressure determined during the flowback period ([Pichon et al. 2018](#), [Liu and Economides 2022](#), [González and Arhancet 2022](#)) [Wang and Sharma \(2020\)](#) proposed applying pressure-transient analysis methods to formation linear flow data (if present), which occurs after the maximum pressure is reached, to estimate pore pressure. However, in that study, reservoir pressure was determined for only one field example, and no independent reservoir pressure measurement was available to validate the estimates derived from linear-flow analysis.

This has motivated the authors to systematically investigate shut-in data (often referred to as rebound data) recorded after DFIT-FBA tests, where an estimate of reservoir pressure is also available from flowback period analysis. This will then be used to determine the viability of obtaining pore pressure estimates from the flowback data. For cases where pore pressures can be obtained from both the flowback and shut-in periods, the values will be compared to evaluate their relationship.

Theory and Methods

A DFIT-FBA test with an extensive post-flowback shut-in was simulated using the ResFrac® software. The boundaries of the reservoir were set far enough from the well to ensure that the fracture does not hit the reservoir boundaries ([Figure 2](#)). The input parameters utilized to initialize the single-phase simulation model are summarized in [Table 1](#). In this section, the simulation model output is used to study mechanisms occurring during the shut-in period. In the next section, the model output is used to illustrate a customized workflow for analyzing the shut-in data for pore pressure.

Table 1—Input parameters used in the simulation case.

Input parameter	Value	Input parameter	Value
Permeability, md	2.03e-4	Total compressibility, 1/psi	1.3e-5
Porosity, fraction	0.05	Measured depth, ft	17060
Reservoir pressure, psi	4351	True vertical depth, ft	7218
Minimum in-situ stress, psi	6527	Well inner diameter, in	5
Young's modulus, psi	4.64e6	Injection rate, bbl/min	5.50
Poisson's ratio	0.25	Flowback rate, bbl/min	0.55
Reservoir fluid viscosity, cp	0.3	Fracture geometry	Radial

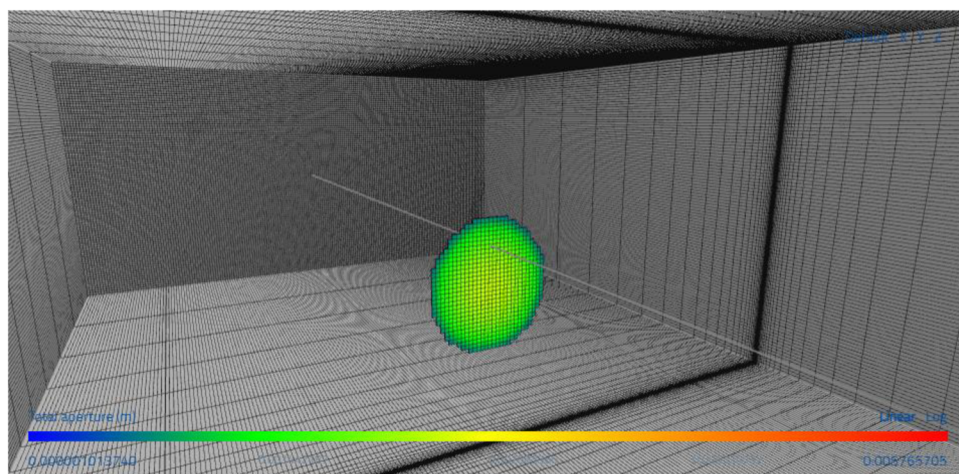


Figure 2—The reservoir boundaries were positioned sufficiently far from the well to prevent the fracture from reaching them.

The bottomhole pressure resulting from this simulation is depicted in Figure 3a. The pressure rises to a maximum pressure and decreases gradually afterwards. As shown in Figure 3b, except for the short period at the start of the shut-in, the post-flowback shut-in pressure is above the input reservoir pressure for the rest of the simulated test.

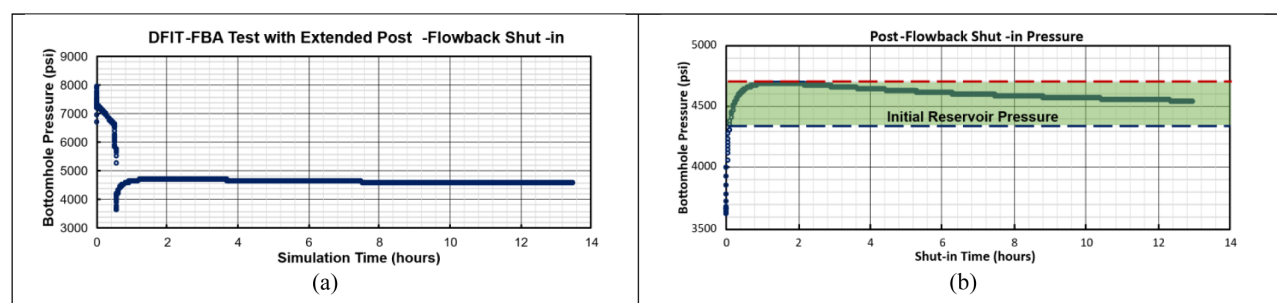


Figure 3—Simulated DFIT-FBA test followed by an extensive shut-in period after the flowback period. (a) Bottomhole pressure for the entire test sequence. (b) Focus on the post-flowback shut-in period. Note that shut-in pressures increase to a maximum then decrease. Except for the very short period at the start of the shut-in, the pressure is above the initial reservoir pressure.

A flow-regime identification plot (pressure drop, dP , and its semi-log derivative, dP' , vs. shut-in time) is presented in Figure 4a. A sudden decrease in the derivative is observed, resembling the signature typically seen in bounded reservoirs. However, as previously explained, the reservoir boundaries were defined in such a way that the pressure signature cannot reflect any boundary effects, given the low permeability used in the model (200 nd). Upon closer inspection of the dP' graph in Figure 4a, there is no derivative data for the corresponding dP values in the orange region. This absence is due to the unique behavior of the DFIT-FBA post flowback shut-in, which has not been observed for conventional well testing previously: the occurrence of negative pressure derivatives as a result of a maximum pressure point (MP). Figure 4b presents the same data on a semi-logarithmic scale, allowing for a clearer view of the negative dP' values after MP.

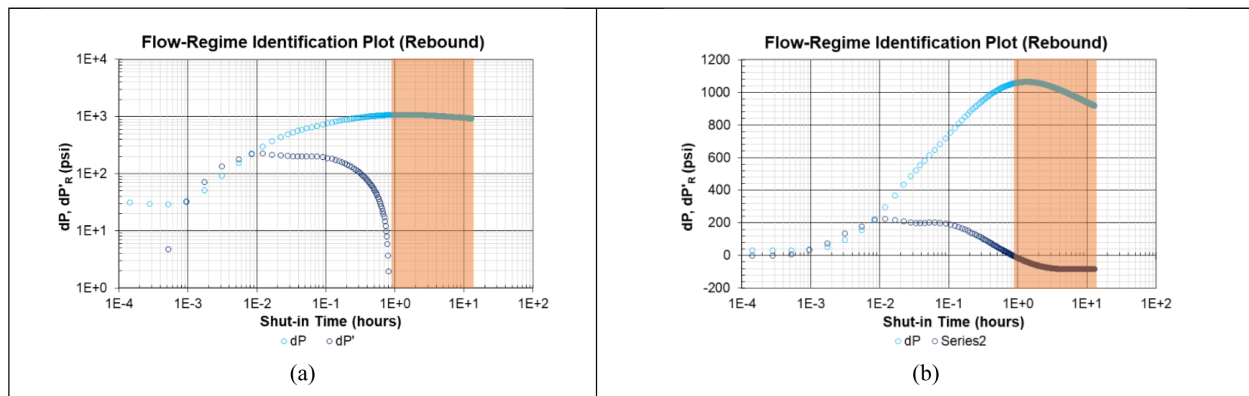


Figure 4—(a) Flow-regime identification plot (log-log plot of dP and dP' versus shut-in time). The late-time derivative signature resembles a bounded reservoir. There is no data for dP' in the orange region. (b) The same data plotted using a Cartesian y-axis allows negative values of dP' to be observed, a unique signature of DFIT-FBA post-flowback shut-in pressure.

This unique derivative signature in Figure 3 can be explained as follows. Figure 5a illustrates the pressure profile in the reservoir, perpendicular to a fracture created during a DFIT-FBA test, at the end of the injection period. During the flowback period, two mechanisms control the pressure in the fracture: flowback and leakoff. If leakoff occurs quickly (high reservoir permeability), the pressure within the fracture decreases uniformly, leading to near-simultaneous closure along the entire length of the fracture. For lower reservoir permeability cases, the flowback rate dominates the pressure response, leading to a greater pressure drop near the wellbore. As a result, the fracture closes more quickly in this region and gradually loses connectivity with the wellbore. However, due to asperities on the fracture walls, the fracture retains some connectivity with the wellbore. If this level of connectivity is not sufficient to support the pressure drop caused by flowback, well unloading will dominate the pressure response. In this case, sections of the fracture further away from the wellbore would not contribute to flowback, and as a result, may remain open (i.e., pressure in the open fracture is greater than the minimum in-situ stress) at the end of the flowback period, while pressure in the wellbore is significantly below reservoir pressure. This scenario is illustrated in Figure 5b. Additionally, a pressure hump develops in the reservoir pressure profile due to low reservoir permeability and a relatively short flowback period (typically under 3 hours), preventing the pressure pulse from dissipating in the reservoir.

As the post-flowback shut-in phase begins, because the permeability in the fracture is still higher than that in the reservoir, flow from the open ends of the fracture occurs toward the closed section rather than into the reservoir. This causes the pressure in the closed section to rise rapidly. As a result, flow occurs from the fracture surface into the reservoir, similar to what happens during the injection phase (Figure 5c). This continues until the pressure in the open fracture sections drops below the minimum in-situ stress, and the closure process is completed throughout the fracture. At this point, the fracture pressure is higher than the initial reservoir pressure but still lower than the current average pressure of the fluid bank region (FBR, next to the fracture surfaces). Consequently, the flow reverses from the reservoir toward the fracture surface, causing the fracture pressure, and by extension, the bottomhole pressure, to decrease again (Figure 5d).

This signature is somewhat analogous to an injection/falloff test, where the increasing pressure before the maximum point (MP) can be compared to the injection phase, and the following decreasing pressure during the falloff period. Although this analogy is not perfect due to the differing pressure slopes, it highlights the fact that, during the DFIT-FBA post-flowback shut-in, the direction of flow in the reservoir changes. Therefore, an analogous approach to injection/falloff test analysis can be used for DFIT-FBA post-flowback shut-in analysis. Specifically, if the post-flowback shut-in data is sufficiently long to pass the MP, the pressure data before the MP can be disregarded, and only the pressures after the MP should be analyzed.

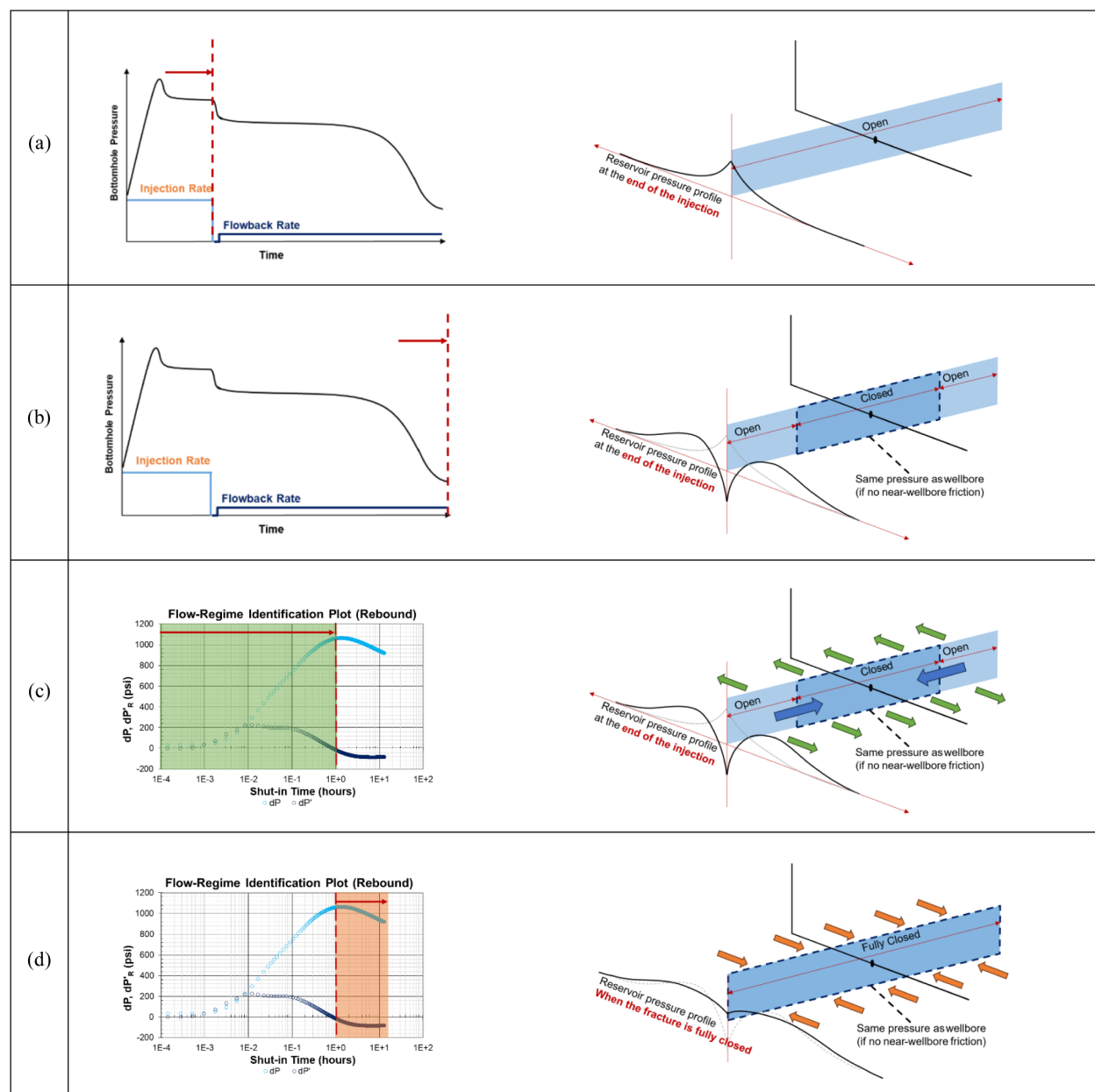


Figure 5—(a) Pressure profile in the reservoir at the end of the injection phase. (b) Pressure profile in the reservoir at the end of the flowback phase. The maximum pressure drop near the wellbore causes early fracture closure in this region, leaving sections of the fracture open at the end of flowback. A hump forms in the reservoir pressure profile due to low reservoir permeability preventing pressure dissipation. (c) During the post-flowback shut-in phase, the higher fracture permeability enables flow from open to closed sections of the fracture, rapidly increasing pressure in the closed section and driving flow into the reservoir until fracture pressure drops below the minimum in-situ stress. (d) When the fracture pressure drops below average FBR pressure, a reversal in flow, from the reservoir to the fracture, occurs.

Results and Discussion

Simulation of Post-Flowback Shut-in Period

Using the conceptual model explained in the Theory and Methods section, a numerical simulation model, also described in that section, was constructed to enable analysis of the bottomhole pressure during the post-flowback shut-in. The results of this sub-section will be used to guide the analysis of field cases provided in the next sub-section.

Figure 6a provides a flow-regime identification plot that exhibits a negative half-slope on the derivative at a shut-in time of 0.4 hours. At this point, the pressure is still rising and has not yet reached the MP. To

determine if this signature corresponds to a formation impulse linear flow, a linear flow-regime identification plot (dP and dP' versus $\Delta t^{0.5}$) is also applied to the same data. Figure 6b demonstrates that linear flow (zero slope line) has not developed, and thus, the result should be interpreted with caution.

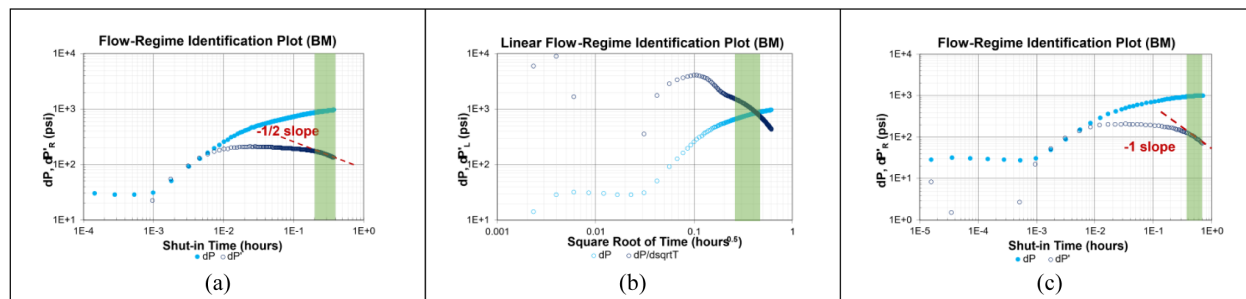


Figure 6—Flow-regime identification applied to simulated post-flowback shut-in results. Data analyzed occurs before the maximum pressure (MP). (a) A negative half-slope forms after $\Delta t_{\text{shut-in}} = 0.4$ hours, suggesting impulse linear flow. (b) Linear flow regime identification plot (dP and dP' vs $\Delta t_{\text{shut-in}}^{0.5}$) applied to the same data as (a); this plot does not confirm that linear flow is occurring in the reservoir. (c) A negative unit-slope line forms after $\Delta t_{\text{shut-in}} = 0.7$ hours, which is the signature of impulse radial flow; this is a false radial flow period and should not be used for analysis.

Extending the data to a shut-in time of 0.7 hours, the flow-regime identification plot (Figure 6c) displays a negative unit slope, which is a characteristic of impulse radial flow. However, this is believed to be a false radial flow signature. The low permeability of the reservoir prevents rapid dissipation of pressure pulses. The depth of investigation barely extends beyond a few meters in such a short time, where the final fracture radius is 15 m. It will be shown later that a linear flow signature will appear in the data, invalidating any preceding radial flow signatures.

The pressure reaches the MP after 1.36 hours of falloff. To analyze the pressure data beyond this point, the falloff time is reset ($\Delta t = 0$ at MP). After 2.44 hours from the end of the flowback period (1.08 hours after MP), another unit slope trend is observed (Figure 7a). For the same reason as described above, this false impulse radial flow should be disregarded; however, an analysis has been performed on both false radial flow periods and will be reported on later. A half-slope line, indicative of formation impulse linear flow, is observed 2.42 hours after the MP (Figure 7b). After this point, no further flow regime signatures are noted. The total shut-in time was 13 hours. It is important to note that in real cases, the test duration required to reach these flow regimes would likely be longer, as the simulation model assumes homogeneous reservoir and fluid behavior for simplicity.

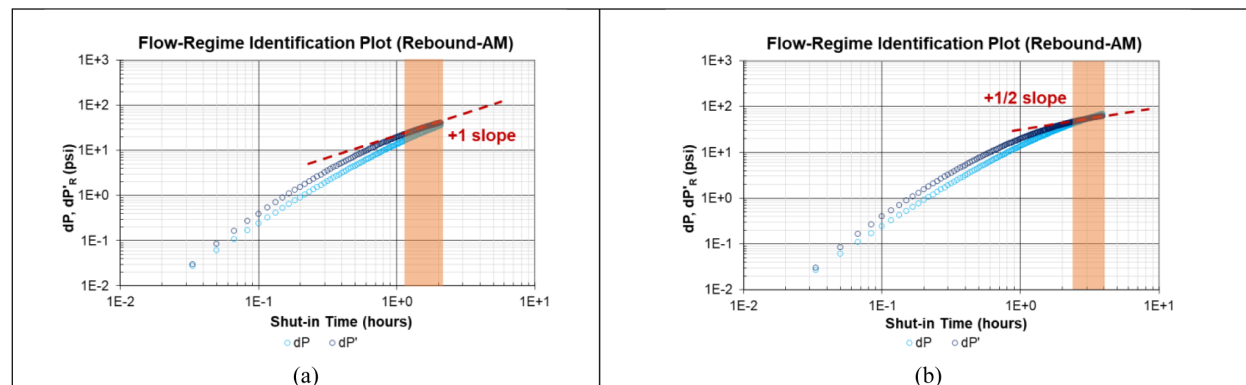


Figure 7—Flow-regime identification applied to simulated post-flowback shut-in results. Data analyzed occurs after the maximum pressure (MP). The shut-in time is reset at MP ($\Delta t_{\text{shut-in}} = 2.0$ hours $\rightarrow \Delta t_{\text{MP}} = 0.0$ hours) (a) Another false impulse radial flow signature is observed after $\Delta t_{\text{shut-in}} = 3.6$ hours ($\Delta t_{\text{MP}} = 1.6$ hours) (c) With further shut-in time, a half-slope line is observed, which is the signature for impulse linear flow.

The estimated reservoir pressures, based on the described flow regimes, are summarized in Table 2. A key observation is that, even after an extended shut-in, the estimated pressure remains above the initial reservoir pressure. This can be attributed to the elevated pressure near the fracture due to the injection period, as well as the low reservoir permeability, which prevents rapid dissipation of the pressure pulses, confirming that the pressure pulse has not propagated beyond the FBR width. It also emphasizes that radial flow analysis should be used with caution, only if the test duration allows the pressure pulse to reach a distance at least equal to the fracture radius.

Table 2—Estimation of reservoir pressure using different flow regimes occurring before and after the maximum pressure achieved during a simulated DFIT-FBA post-flowback shut-in period.

	Unit	Linear	Radial	Radial	Linear
AM/BM¹		BM	BM	AM	AM
Reservoir Pressure (Shut-in Analysis)	psi	4878	4747	4595	4474
Reservoir Pressure (Model input)	psi	4351	4351	4351	4351
Error	%	12.1%	9.1%	5.6%	2.8%

¹ AM: After Maximum, BM: Before Maximum

Based on the results from this simulation study, a recommended procedure to analyze the DFIT-FBA post-flowback shut-in data is provided in Figure 8.

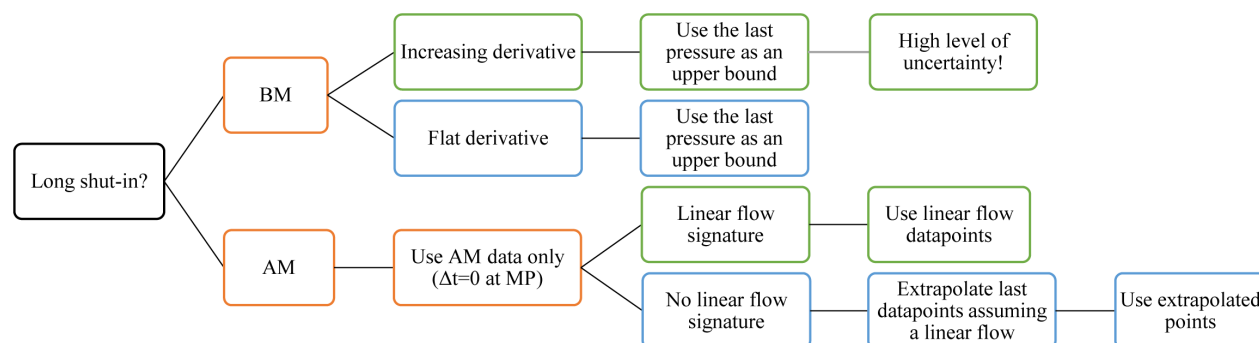


Figure 8—Proposed procedure to analyze DFIT-FBA post-flowback shut-in test data.

Field Cases

Using the procedure in Figure 8, DFIT-FBA field cases (FC) with post-flowback shut-in data were analyzed for reservoir pressure. All field cases are from North American (US and Canada) unconventional reservoirs, and the shut-in times range from 5 to 105 hours. Table 3 summarizes the test information. Minimum in-situ stress (S_{min}) and reservoir pressure were derived from the analysis of the flowback cycle of the DFIT-FBA tests. The results of the post-flowback shut-in analysis are summarized in Table 4. The reader is referred to Appendix A for additional information and plots of the test data. Highlights of the field case analyses are summarized as follows:

Table 3—Closure and reservoir pressure estimates from the flowback period of DFIT-FBA field cases.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#1	6106	3819
FC#2	6005	4322
FC#3	5656	3814
FC#4	7179	5236
FC#5	Not Available ¹	Not Available ¹
FC#6	8311	6237
FC#7	7324	4150
FC#8	6976	3539
FC#9	5903	4540

¹ Failed test**Table 4—Results of post-flowback shut-in data analysis using the workflow in Figure 8.**

	Unit	FC#1	FC#2	FC#3	FC#4	FC#5	FC#6	FC#7	FC#8	FC#9
BM/AM	-	AM	AM	AM	AM	AM	AM	AM	BM	BM
S_{min}	psi	6106	6005	5656 ^{III}	7179	NA	8311	7324	6976	5903 ^{III}
Maximum BHP	psi	6106 ^{IV}	6005	4134 ^{III}	6744	7223	7658	5802	4366	4315 ^{III}
Shut-in time to maximum BHP	hr	0.8	2.8	12.2 ^{III}	8.4	6.9	43.6 ^{II}	5.5 ^V	15.8	4.8 ^{III}
Final shut-in time	hr	80.6	81.9	63.3	18.2	24.0	105.4	5.7 ^V	15.8	4.8
Flow regime (L/LE ¹)	-	L	L	L	L	L	L	LE	-	-
Reservoir Pressure (Rebound)	psi	3559	4403	4042	6656	7156	7614	5629	4366 ^{VI}	4315 ^{VI}
Reservoir Pressure (Flowback)	psi	3819	4322	3814	5236	NA	6237	4150	3539	4540
Reservoir Pressure Difference	%	7.3 ^V	-1.8 ^V	-5.6 ^V	-21.3 ^I	-	-18.1 ^I	-26.3 ^V	-19.1	5.2
Difference (S_{min} & Res. Pressure)	psi	2547	1601	1614	523	-	697	1695	2610	1588
Fraction of S_{min}	%	41.7 ^V	26.7 ^V	28.5 ^V	7.3	-	8.4	23.1 ^V	37.4	26.9 ^V

¹ Linear/Linear Extrapolation

I, II, III, IV, V Correspond to the discussion section

^{VI} Upper bound for reservoir pressure

Cases Where Closure and Reservoir Pressures Are Similar in Magnitude (FC#4 and FC#6).. Excluding the short FC#7 test (with only 5.5 hours of shut-in data), the largest discrepancies between reservoir pressure obtained from flowback analysis, and post-flowback shut-in tests (which were long enough to reach maximum pressure - MP), were observed for FC#4 and FC#6, at -21.3% and -18.1%, respectively. Interestingly, these tests also exhibited closure and reservoir pressures that were close in magnitude, with differences of 7.3% and 8.4%, respectively. As noted previously, Zanganeh et al. (2020a) recommended estimating reservoir pressure during the flowback portion of DFIT-FBA by identifying the start of the second unit slope line on a log-log plot of RNP' versus material balance time (MBT). However, closure signatures can delay the development of this slope, leading to an underestimation of reservoir pressure. Because

reservoir pressure is required to determine permeability from before-closure DFIT-FBA data, inaccuracies can have significant implications. For example, replacing the underestimated reservoir pressure from flowback analysis with rebound-derived values resulted in an order-of-magnitude change in permeability estimates for both FC#4 and FC#6.

Extended Monitoring and Implications for Rebound Pressure (FC#6).. The most delayed MP occurred for the FC#6 test (after 43.6 hours of shut-in), despite the original plan to monitor rebound pressure for just 24 hours. An accidental extension of the monitoring period to 105 hours allowed the MP to be reached. This unexpected result suggests that, given sufficient monitoring time, all DFIT-FBA post-flowback shut-ins will eventually reach MP. Consequently, the procedure for analyzing post-flowback shut-in data has been revised. Previously, if the post-flowback shut-in phase ended before reaching MP (i.e., while the derivative was still increasing), it was suggested to extrapolate the final pressure points, assuming linear flow would develop after the test. However, as the pressure trend is increasing before MP, such extrapolations can overestimate reservoir pressure. A more reliable approach is to treat the last recorded pressure as an approximate upper bound for reservoir pressure, provided the data shows a relatively flat derivative and the monitoring period is adequately extended.

Impact of the Duration of Monitoring and S_{min} Proximity (FC #3 and FC#9).. Analysis of FC#9 led to the hypothesis that post-flowback shut-in pressure might not reach MP when the highest recorded pressure is significantly below S_{min} . However, this assumption was contradicted by results from FC#3, where the difference between the highest post-flowback shut-in pressure and S_{min} was comparable to FC#9. Extending the monitoring period for FC#3 to 12.2 hours (compared to 4.8 hours for FC#9) enabled the post-flowback shut-in data to reach MP and develop linear flow in the reservoir. These findings underscore the importance of prolonged pressure monitoring for accurate MP determination.

Impact of Insufficient Flowback Duration on Rebound Pressure (FC#1).. Rebound pressure can reach/exceed the minimum in-situ stress (S_{min}) if the flowback period is too short to allow adequate drainage of the fracture. This phenomenon was observed for FC#1, where the post-flowback shut-in pressure rose to 6106 psi after just 0.8 hours of flowback. Due to the flowback process concluding after one hour, the average fracture pressure remained above S_{min} at the end of the flowback period. It should be noted that the minimum in-situ stress and reservoir pressure could be estimated from the flowback analysis probably due to partial closure of the fracture near the wellbore.

Characteristics of Tests with Accurate Reservoir Pressure Estimates (FC#1, FC#2, FC#3).. The most accurate alignment between reservoir pressure estimates from flowback and post-flowback shut-in data was observed for FC#1, FC#2, and FC#3. These tests shared two key characteristics: a substantial difference between reservoir pressure and minimum in-situ stress, and sufficient post-flowback shut-in monitoring to reach and pass the MP, allowing transition to linear flow in the reservoir. Similar outcomes might have been possible for FC#7 and FC#9, as both exhibited the first characteristic. However, FC#9 failed to reach MP, while FC#7 reached MP but lacked sufficient post-MP data (only 0.2 hours).

Reassessing Reservoir Pressure Estimates from Flowback.. Previously, it was believed that reservoir pressure estimated during the flowback period, specifically at the start of the second unit slope line, tended to be overestimated. However, a comparison with post-flowback shut-in analysis revealed that this is not the case for most of the field examples analyzed here.

Summary and Conclusions

This study used simulation to analyze a full spectrum of pressure signatures observed during the post-flowback shut-in phase of a DFIT-FBA test. Additionally, an analysis workflow was developed for

application to DFIT-FBA post-flowback shut-in tests and utilized to evaluate nine field cases. The following are the primary conclusions drawn from this study:

- Linear flow can be developed in an unconventional reservoir during a DFIT-FBA post-flowback shut-in test; also, false radial flow signatures may appear before this linear flow regime develops and should not be interpreted for reservoir pressure.
- Rebound reservoir pressure estimates will be generally higher than the actual reservoir pressure, even after an extended shut-in period.
- DFIT-FBA post-flowback shut-in pressures will pass through a maximum if pressure is monitored long enough, as a result of:
 1. Partially closed fractures at the end of the DFIT-FBA flowback phase
 2. Pressure gradient in the reservoir, which is affected by (a) reversal of flow directions during the injection and flowback phases of a DFIT-FBA test, and (b) low permeability of unconventional reservoirs.
- A more accurate reservoir pressure estimate can be achieved if the test design ensures significant data is collected after the maximum point.
- If minimum horizontal stress and reservoir pressure are close, the reservoir pressure signature during flowback might be masked by fracture closure, and the reservoir estimate should be used with caution unless confirmed by additional data, such as post-flowback shut-in information.
- Reservoir pressure estimates can significantly affect before-closure permeability estimates; when reservoir pressure estimates from flowback analysis are uncertain, it is recommended to perform post-flowback shut-in tests, following the guidance in this work

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Nomenclature

c_t	Total compressibility, 1/psi
dP	Pressure steps, psi
dP'_R	Radial pressure derivative, psi/hr
dP'_L	Linear pressure derivative, psi/hr ^{0.5}
k	Reservoir permeability, md
p_{ref}	Reference pressure, psi
p_f	Fracture pressure, psi
q_{fb}	Flowback rate, bbl/hr
r_f	Fracture radius, ft
S_{min}	Minimum in-situ stress, psi
t_{MB}	Material balance time, s
t_D	Dimensionless time

- V_{fb} Cumulative flowback volume, bbl
 ϕ Porosity, fraction
 μ Reservoir fluid viscosity, cp
 Δt Time from start of flowback, hr

Abbreviations

- AM After maximum
 BHP Bottomhole pressure
 BM Before maximum
 DFIT Diagnostic fracture injection test
 DFIT-FBA Diagnostic fracture injection test-flowback analysis
 FBR Fluid bank region
 FC Field case
 L Linear
 LE Linear extrapolation
 MBT Material balance time
 MP Maximum point
 RNP Rate-normalized pressure
 RNP' Rate-normalized pressure derivative
 RTA Rate transient analysis

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Appendix A

Field Case #1

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 6106 and 3819 psi, respectively.

Table A1—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#1	6106	3819

The operator stopped the flowback period after one hour. As a result, the pressure increased sharply to the minimum in-situ stress at the start of shut-in [Figure A1(a)] and decreased afterward. Figures A1(b) and A1(c) provide the flow-regime identification plots. Figure A1(c) is only provided for this field case to show the negative derivatives as an example. Following the procedure described in Figure 8, the BM period is ignored, and the flow-regime identification plot was generated for the AM section [Figure A1(d)]. A clear half slope can be identified; therefore, reservoir pressure was estimated to be 3559 psi versus 3819 psi obtained from the analysis of the flowback period.

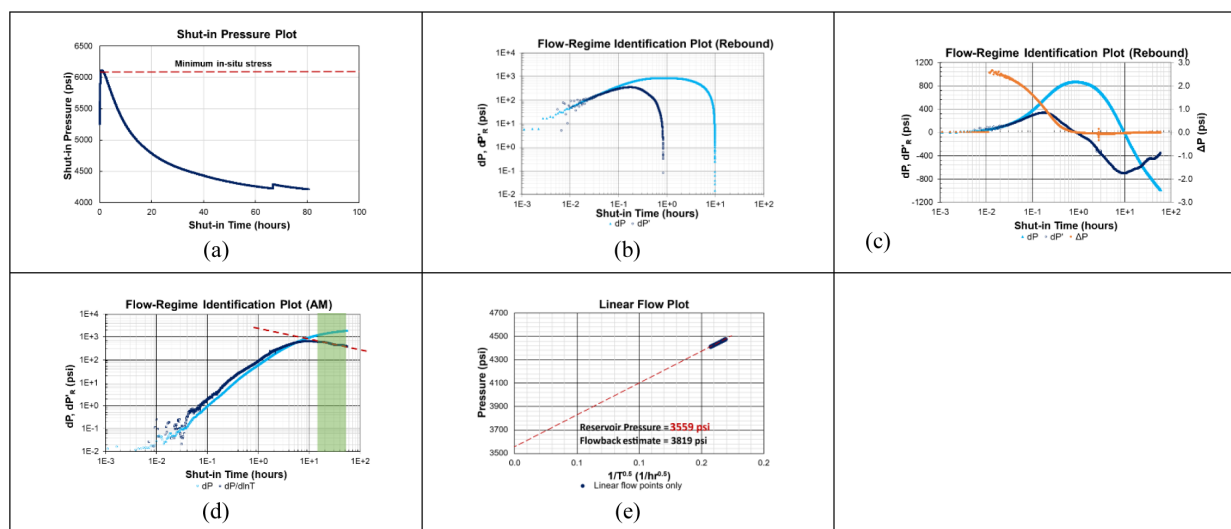


Figure A1—(a) Post flowback shut-in pressure profile. (b) Flow-regime identification (log-log) plot. (c) Flow-regime identification plot with Cartesian y-axis used to highlight the negative pressure derivatives. (d) Flow-regime identification plot for the AM period. (e) Linear flow plot used to estimate reservoir pressure.

Field Case #2

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 6005 and 4322 psi, respectively.

Table A2—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#2	6005	4322

This field case was operated by the same company as FC#1 and presents the same issue. Due to the short flowback period, pressure increased sharply to the minimum in-situ stress at the beginning of shut-in and

then decreased, as shown in Figure A2(a). Figures A2(b) and A2(c) provide the flow-regime identification plots for the entire shut-in and the AM periods, respectively. The reservoir pressure was estimated to be 4403 psi using the linear flow plot [Figure A2(d)], compared to 4403 psi obtained from the flowback period.

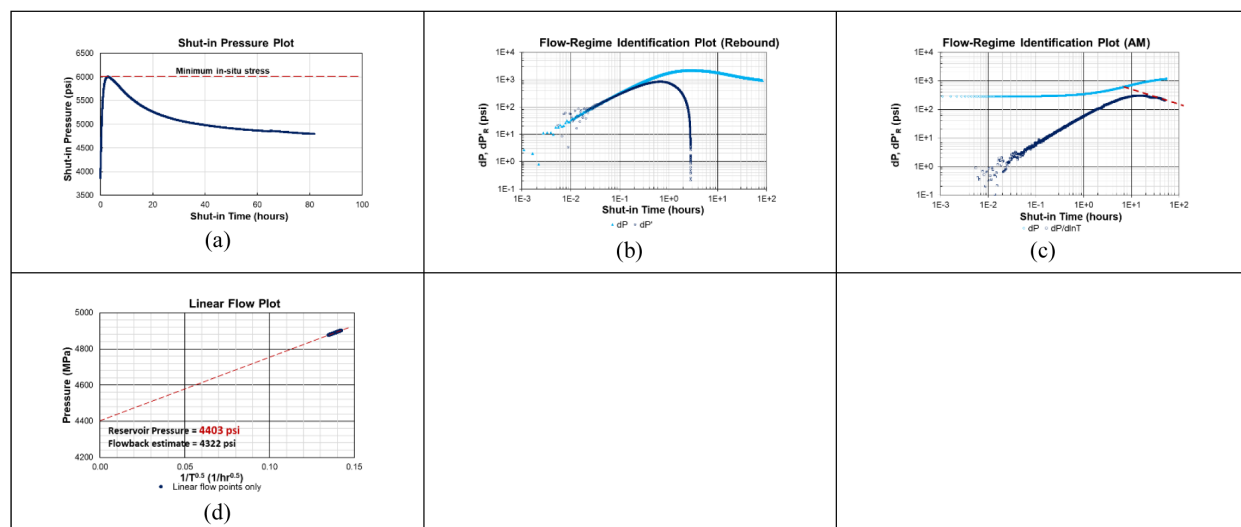


Figure A2—(a) Post flowback shut-in pressure profile. (b) Flow-regime identification plot. (c) Flow-regime identification plot for the AM period. (d) Linear flow plot used to estimate the reservoir pressure.

Field Case #3

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 5656 and 3814 psi, respectively.

Table A3—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#3	5656	3814

The shut-in pressure reached the MP and then decreased, as shown in Figure A3(b). Figures A3(c) and A3(d) provide the flow-regime identification plots for the entire shut-in and the AM periods, respectively. A linear flow regime was confirmed using the linear flow identification plot in Figure A3(d). The reservoir pressure was estimated to be 4042 psi using the linear flow plot in Figure A3(e), compared to 3814 psi obtained from analyzing the flowback period.

Field Case #4

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 7179 and 5236 psi, respectively.

Table A4—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#4	7179	5236

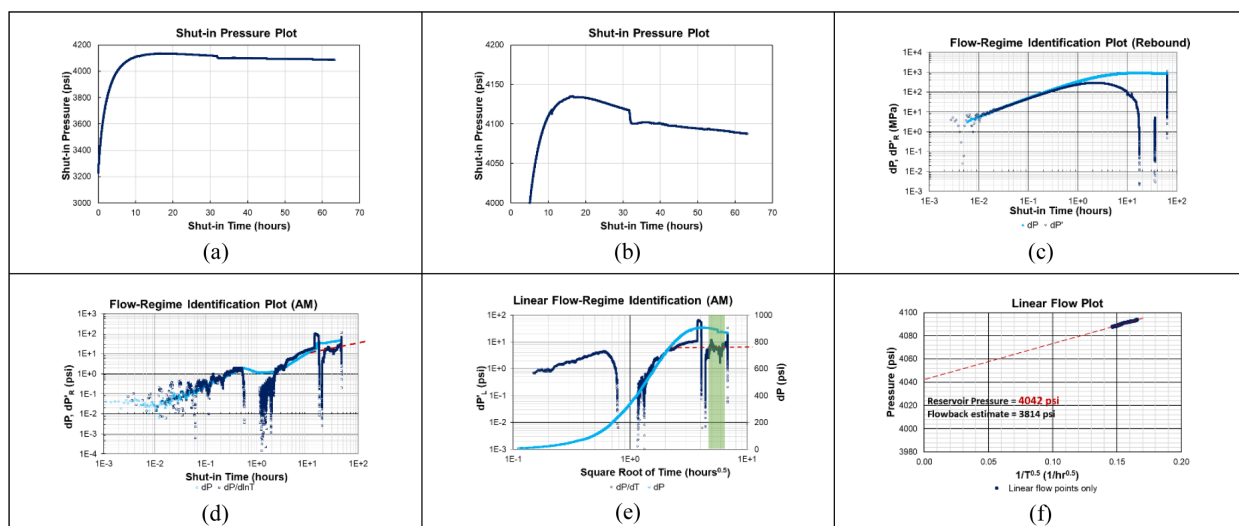


Figure A3—(a) Post flowback shut-in pressure profile. (b) A zoom-in of the pressure profile to highlight the maximum point. (c) Flow-regime identification plot. (d) Flow-regime identification plot for the AM period. (e) Linear flow identification plot for the AM period to confirm the linear flow signature seen in (d). (f) Linear flow plot used to estimate the reservoir pressure.

The shut-in pressure reached the MP and then decreased, as shown in Figure A4(b). Figures A4(c) and A4(d) provide the flow-regime identification plots for the entire shut-in and the AM periods, respectively. A unit slope line appears in the derivative, which is believed to be a false radial flow given its early appearance. Linear extrapolation of the final data points was used to estimate the reservoir pressure, assuming that linear flow will continue to develop in the reservoir. This resulted in a reservoir pressure estimate of 6656 psi, as shown in Figure A4(e), compared to 5236 psi obtained from analyzing the flowback period.

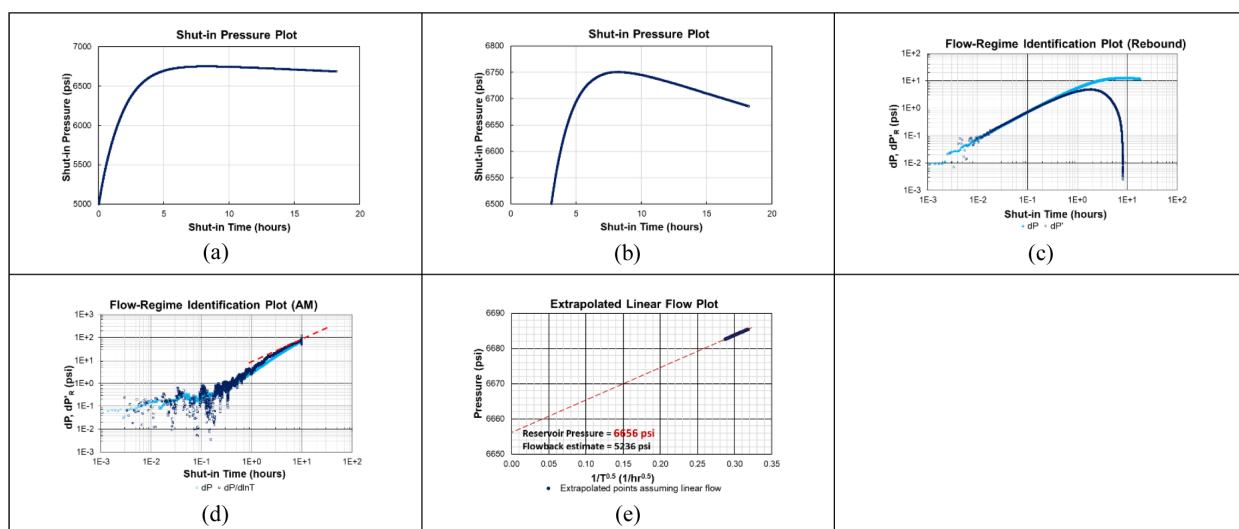


Figure A4—(a) Post flowback shut-in pressure profile. (b) A zoom-in of the pressure profile to highlight the maximum point. (c) Flow-regime identification plot. (d) Flow-regime identification plot for the AM period. (e) Linear extrapolation of the final data points to estimate the reservoir pressure, assuming that linear flow will continue to develop in the reservoir.

Field Case #5

The pressure gauges malfunctioned during the flowback phase, resulting in unusable test data. The operator then pumped another DFIT-FBA test, which is the next case discussed (FC#6).

Table A5—Pressure gauges failed to record the data during the flowback period.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#5	Not Available	Not Available

For this case, however, 24 hours of shut-in data were successfully recorded and used to estimate the reservoir pressure. The shut-in pressure initially reached MP before declining, as depicted in Figure A5(b). Flow-regime identification plots for the full shut-in period and the AM period are shown in Figures A5(c) and A5(d), respectively. A unit slope line is visible in the pressure derivative, and upon closer examination, signs of late linear flow development can also be observed [Figure A5(e)]. Based on the final data points, the reservoir pressure was estimated to be 7156 psi.

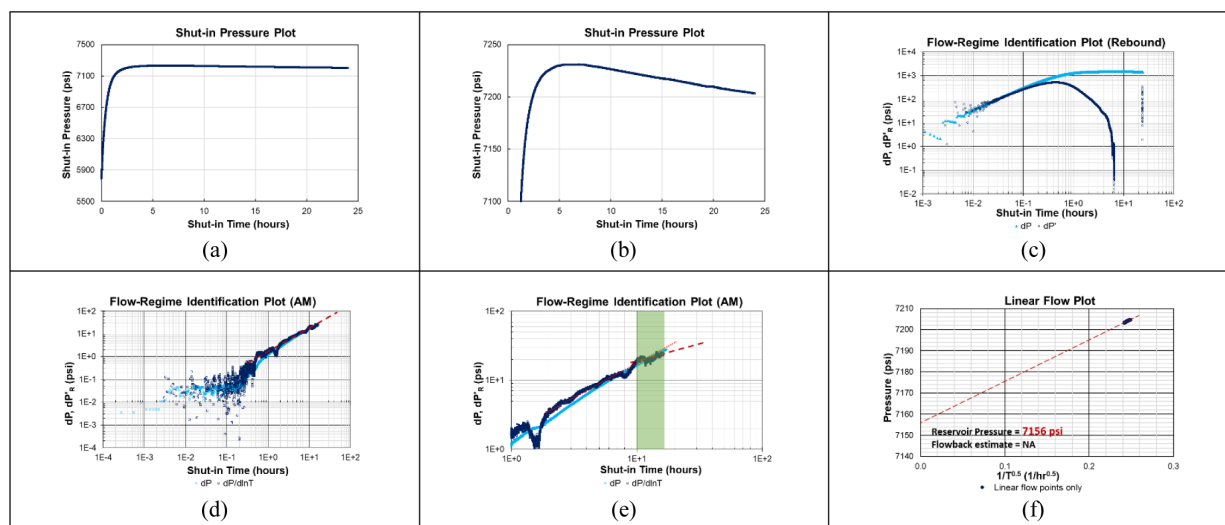


Figure A5—(a) Post flowback shut-in pressure profile. (b) A zoom-in of the pressure profile to highlight the maximum point. (c) Flow-regime identification plot. (d) Flow-regime identification plot for the AM period. (e) A zoom-in of the flow regime identification plot illustrates the development of a linear flow at the end of the test. (f) Linear flow plot used to estimate the reservoir pressure.

Field Case #6

Analysis of the DFIT-FBA test for this case (a repeat test to replace the failed FC#5 flowback test) resulted in minimum in-situ stress and reservoir pressure estimates of 8311 and 6237 psi, respectively.

Table A6—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#6	8311	6237

A 24-hour post-flowback shut-in measurement was originally planned for this test. However, due to an oversight, the downhole gauges were not removed, resulting in an extended 105-hour shut-in period. As shown in Figure A6(b), the MP occurs at 43.6 hours, well beyond the intended shut-in duration. This is also the latest occurrence of the maximum point among all nine field cases analyzed herein, with the second latest being at 12.2 hours. Flow-regime identification plots for the full shut-in period and AM periods are presented in Figures A6(c) and A6(d), respectively. A half-slope line is observed in Figure A6(d), and therefore a linear flow plot was used to estimate reservoir pressure in Figure A6(e). This analysis yields a reservoir pressure of 7614 psi, compared to 6237 psi estimated from the flowback period.

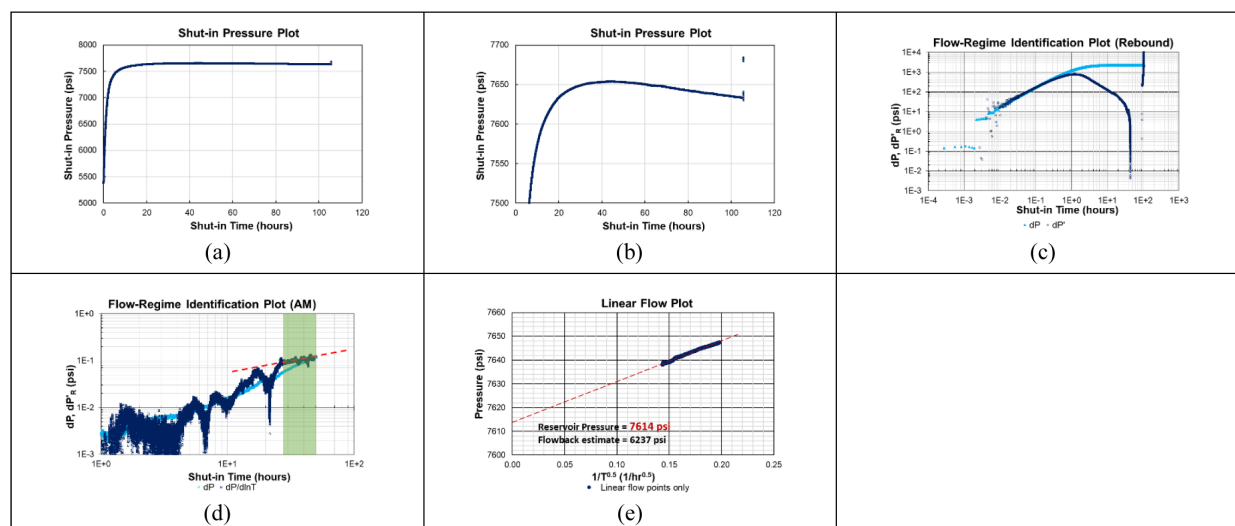


Figure A6—(a) Post flowback shut-in pressure profile. (b) A zoom-in of the pressure profile to highlight the maximum point. (c) Flow-regime identification plot. (d) Flow-regime identification plot for the AM period. (e) Linear flow plot used to estimate the reservoir pressure.

Field Case #7

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 7324 and 4150 psi, respectively.

Table A7—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#7	7324	4150

Although the shut-in pressure initially appeared to increase steadily, closer inspection—shown in Figure A7(b)—revealed that the pressure reached a peak at approximately 5.5 hours before beginning to decline. Flow-regime identification plots for the full shut-in duration and AM periods are presented in Figures A7(c) and A7(d), respectively. Due to limited pressure data acquired during the AM period, no distinct flow regime could be identified from the pressure derivative. As a result, a linear extrapolation of the final pressure points was employed to estimate the reservoir pressure, under the assumption that linear flow continues to evolve. This approach yielded a reservoir pressure of 5629 psi, in contrast to the 4150 psi estimated from the flowback period.

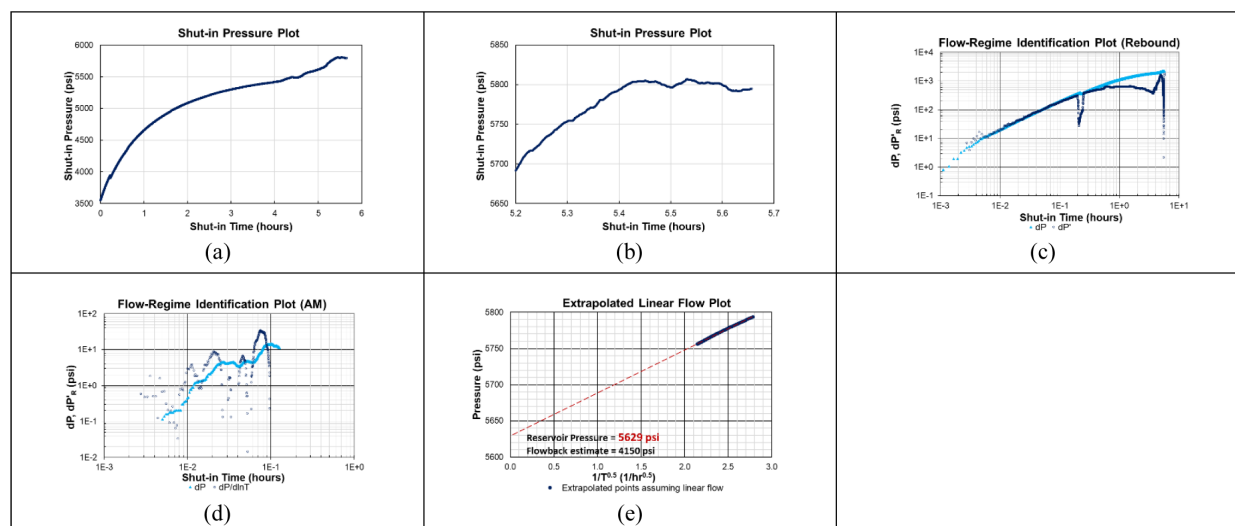


Figure A7—(a) Post flowback shut-in pressure profile. (b) A zoom-in of the pressure profile to highlight the maximum point. (c) Flow-regime identification plot. (d) Flow-regime identification for the AM period. (e) Linear extrapolation of the final data points to estimate the reservoir pressure, assuming that linear flow will continue to develop in the reservoir.

Field Case #8

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 6976 and 3539 psi, respectively.

Table A8—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#8	6976	3539

The post-flowback shut-in pressure increased throughout the entire test. The flow-regime identification plot, shown in Figure A8(b), displays a negative unit slope line. As previously discussed, this is likely a false radial flow signature, due to its early occurrence. As the derivative is still increasing, the customized workflow suggests considering the last recording pressure (4366 psi) as an upper bound for the reservoir pressure; however, this estimate is highly uncertain. Flowback analysis resulted in a reservoir pressure estimate of 3539 psi.

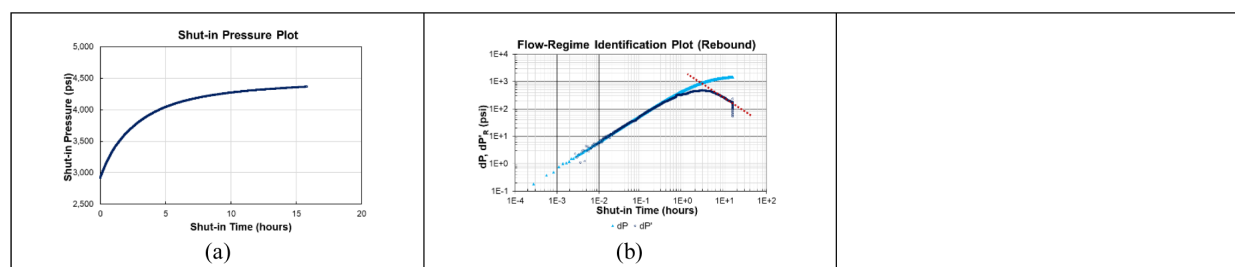


Figure A8—(a) Post flowback shut-in pressure profile. (b) Flow-regime identification plot.

Field Case #9

Analysis of the DFIT-FBA test for this case resulted in minimum in-situ stress and reservoir pressure estimates of 5903 and 4540 psi, respectively.

Table A9—Minimum in-situ stress and reservoir pressure estimates using the flowback data from the flowback portion of the test.

	Minimum in-situ stress (S_{min}) (psi)	Flowback reservoir pressure (psi)
FC#9	5903	4540

Similar to FC#8, the post-flowback shut-in pressure increased throughout the entire test. A negative half-slope line is observed on the flow-regime identification plot [Figure A9(b)]. However, a zero-slope line did not appear on the linear flow regime identification plot [Figure A9(c)]. The derivative is almost flat at the end of the test; therefore, the customized workflow suggests considering the last recording pressure (4366 psi) as an upper bound for reservoir pressure. Flowback analysis resulted in a reservoir pressure estimate of 4540 psi.

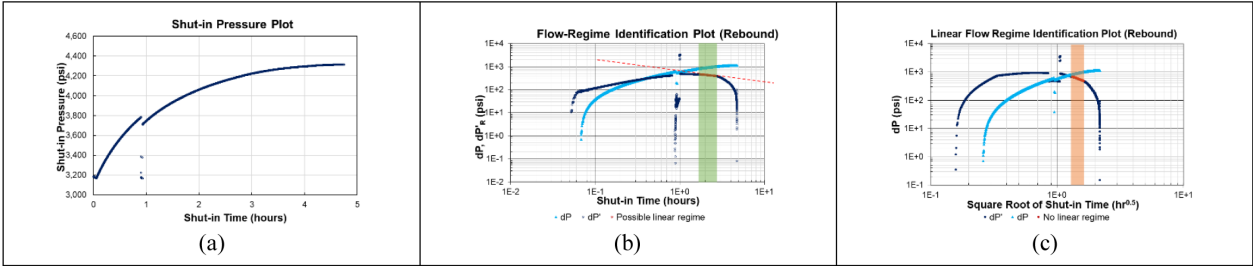


Figure A9—(a) Post flowback shut-in pressure profile. (b) Flow-regime identification plot. (c) Linear flow identification plot used to confirm the linear flow signature observed in (b).